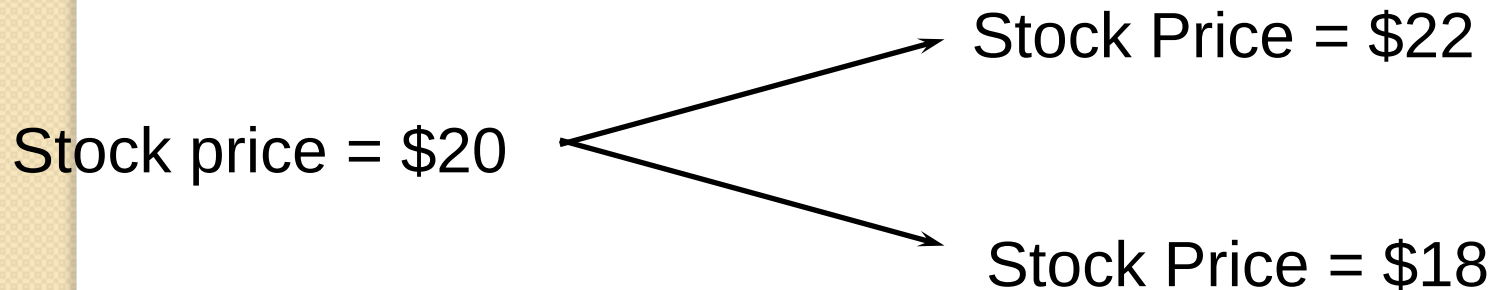




Binomial Option Pricing Models

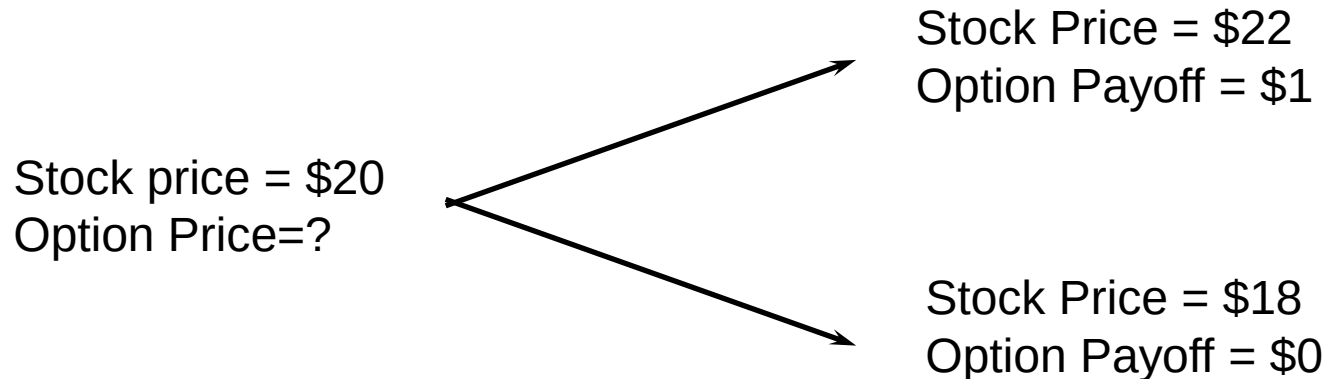
A Simple Binomial Model

- A stock price is currently \$20
- In 3 months it will be either \$22 or \$18



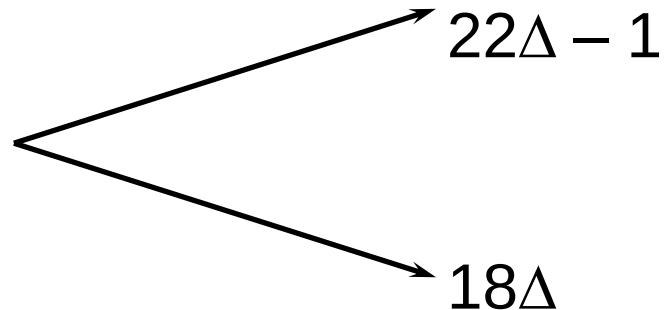
A Call Option

A 3-month call option on the stock has a strike price of 21.



Setting Up a Riskless Portfolio

- For a portfolio that is long Δ shares, and a short 1 call option values are



Portfolio is riskless when $22\Delta - 1 = 18\Delta$ or $\Delta = 0.25$

Valuing the Portfolio

(Risk-Free Rate is 4%)

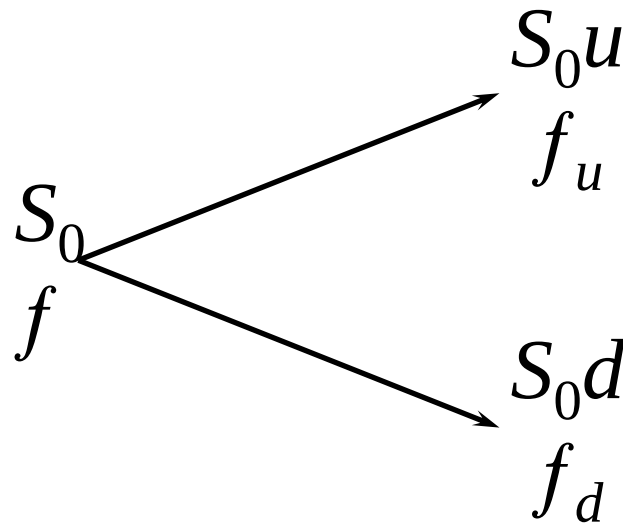
- The riskless portfolio is:
long 0.25 shares
short 1 call option
- The value of the portfolio in 3 months is
 $22 \times 0.25 - 1 = 4.50$
- The value of the portfolio today is
 $4.5e^{-0.04 \times 0.25} = 4.455$

Valuing the Option

- The portfolio that is
 long 0.25 shares
 short 1 option
is worth 4.455
- The value of the shares is
 5.00 ($= 0.25 \times 20$)
- The value of the option is therefore
 $5.00 - 4.455 = 0.545$

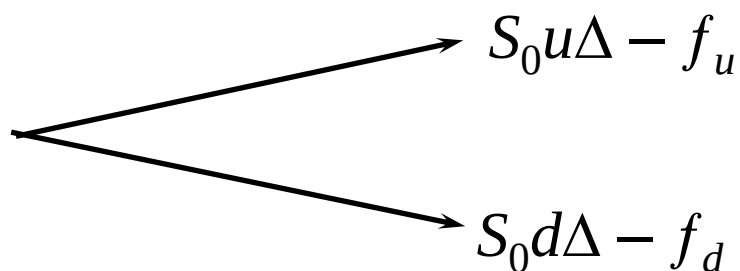
Generalization

A derivative lasts for time T and is dependent on a stock



Generalization

- Value of a portfolio that is long Δ shares and short 1 derivative:



- The portfolio is riskless when $S_0 u \Delta - f_u = S_0 d \Delta - f_d$
or

$$\Delta = \frac{f_u - f_d}{S_0 u - S_0 d}$$

Generalization

- Value of the portfolio at time T is $S_0 u \Delta - f_u$
- Value of the portfolio today is $(S_0 u \Delta - f_u) e^{-rT}$
- Another expression for the portfolio value today is $S_0 \Delta - f$
- Hence

$$f = S_0 \Delta - (S_0 u \Delta - f_u) e^{-rT}$$

Generalization

(continued)

Substituting for Δ we obtain

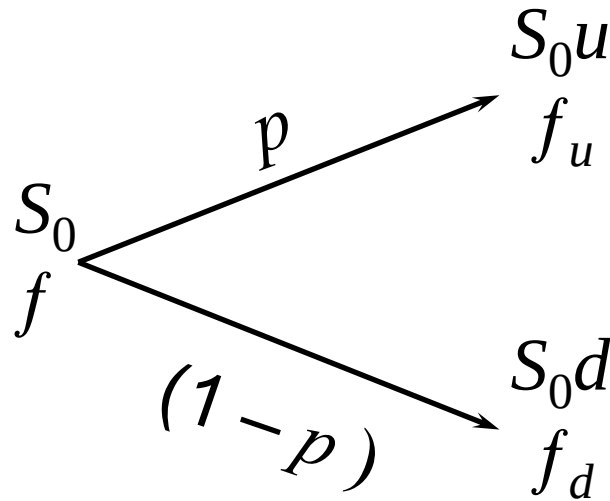
$$f = [pf_u + (1 - p)f_d]e^{-rT}$$

where

$$p = \frac{e^{rT} - d}{u - d}$$

p as a Probability

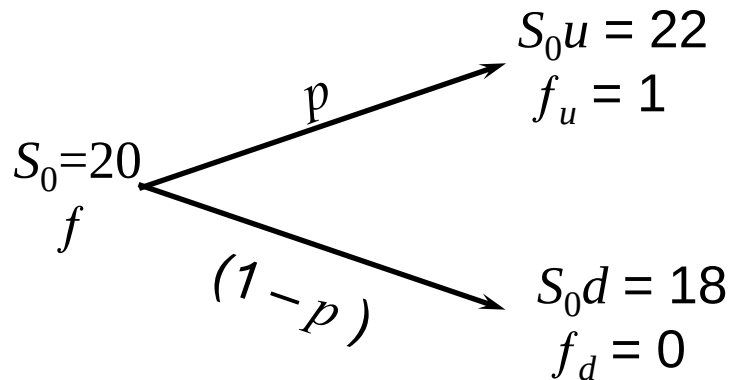
- It is natural to interpret p and $1-p$ as probabilities of up and down movements
- The value of a derivative is then its expected payoff in a risk-neutral world discounted at the risk-free rate



Risk-Neutral Valuation

- When the probability of an up and down movements are p and $1-p$ the expected stock price at time T is $S_0 e^{rT}$
- This shows that the stock price earns the risk-free rate
- Binomial trees illustrate the general result that to value a derivative we can assume that the expected return on the underlying asset is the risk-free rate and discount at the risk-free rate
- This is known as using risk-neutral valuation

Original Example Revisited



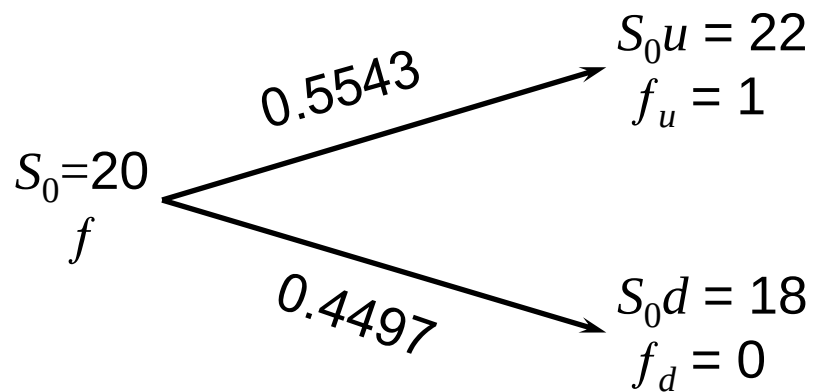
p is the probability that gives a return on the stock equal to the risk-free rate:

$$20e^{0.04 \times 0.25} = 22p + 18(1-p) \text{ so that } p = 0.5503$$

Alternatively:

$$p = \frac{e^{rT} - d}{u - d} = \frac{e^{0.04 \times 0.25} - 0.9}{1.1 - 0.9} = 0.5543$$

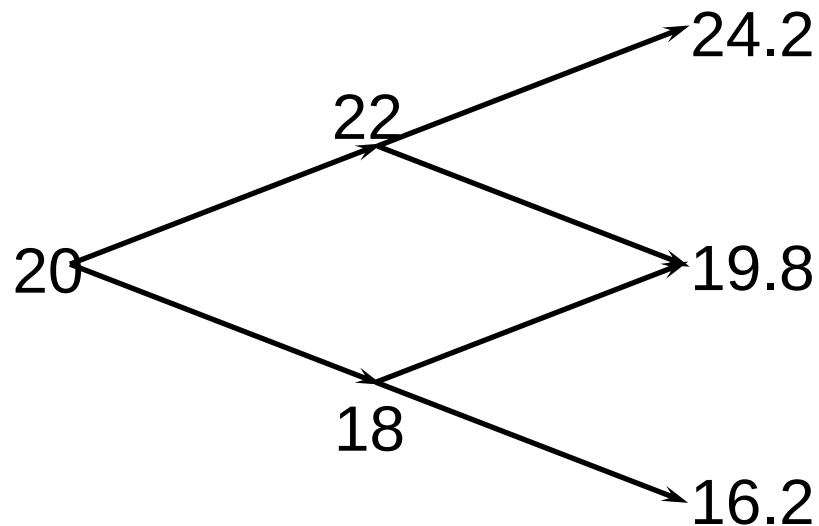
Valuing the Option Using Risk-Neutral Valuation



The value of the option is

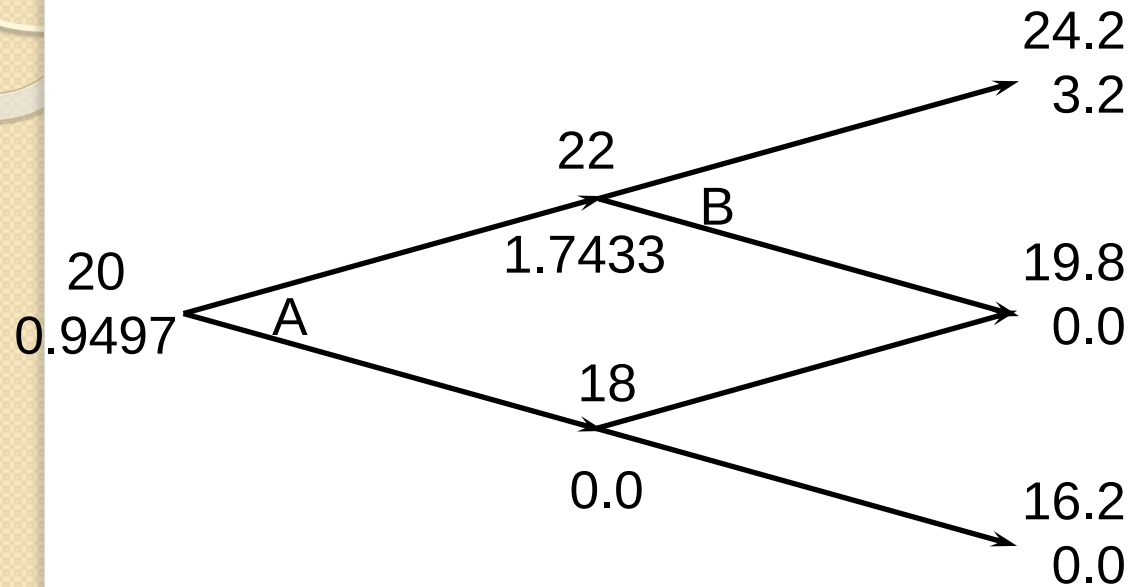
$$e^{-0.04 \times 0.25} (0.5543 \times 1 + 0.4497 \times 0) \\ = 0.545$$

A Two-Step Example



- $K=21$, $r = 4\%$
- Each time step is 3 months

Valuing a Call Option



Value at node B

$$= e^{-0.04 \times 0.25} (0.5503 \times 3.2 + 0.4497 \times 0) = 1.7433$$

Value at node A

$$= e^{-0.04 \times 0.25} (0.5503 \times 1.7433 + 0.4497 \times 0) = 0.9497$$

$$f = e^{-r\Delta t} [P f_u + (1-P) f_d]$$

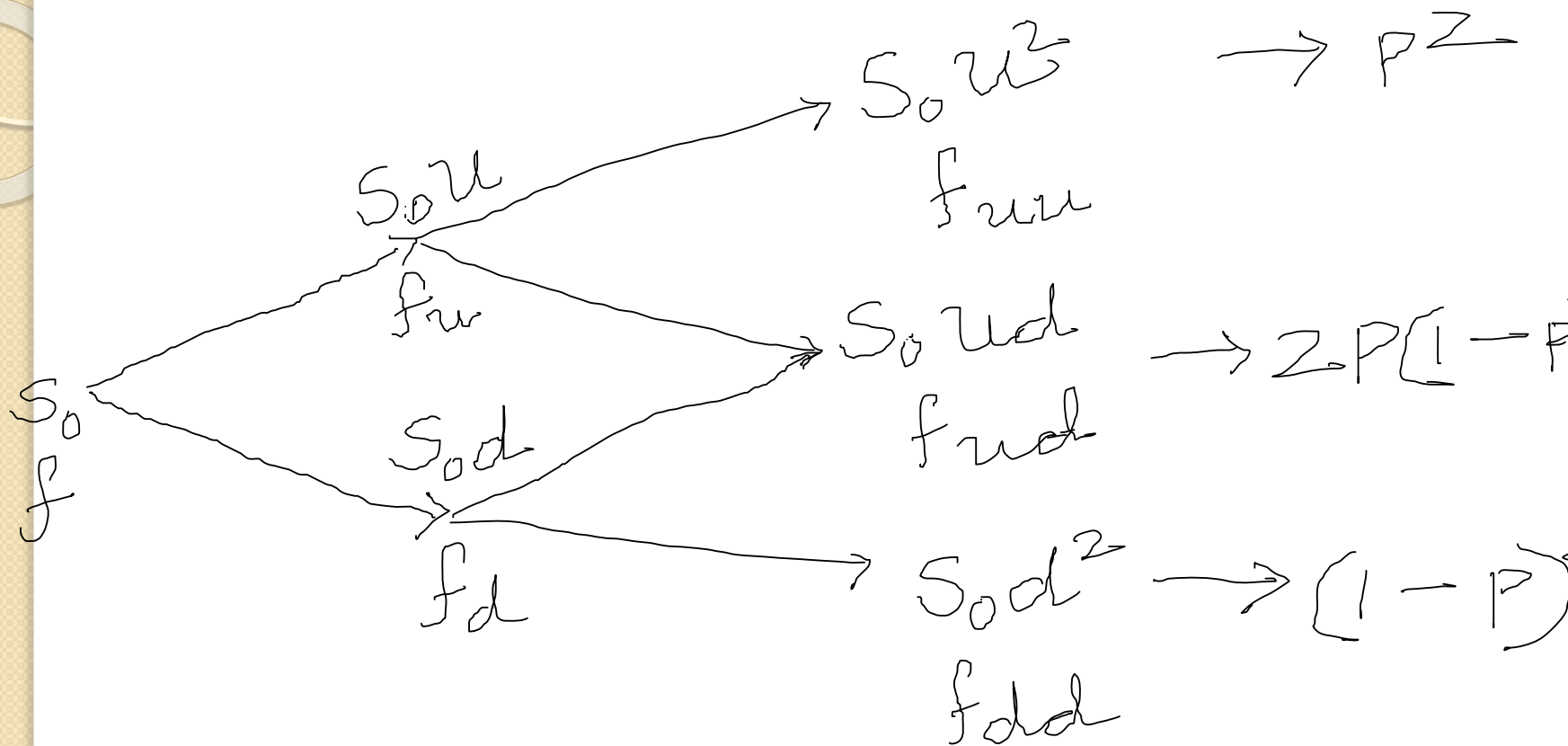
$$\text{where } P = \frac{e^{r\Delta t} - d}{u - d}$$

$$f_u = e^{-r\Delta t} [P f_{uu} + (1-P) f_{ud}]$$

$$f_d = e^{-r\Delta t} [P f_{ud} + (1-P) f_{dd}]$$

$$f = e^{-r\Delta t} [P f_u + (1-P) f_d]$$

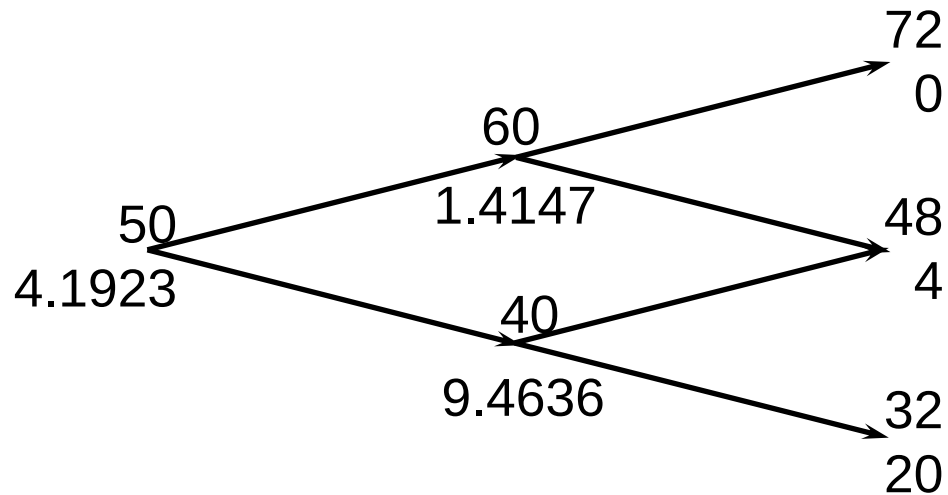
$$f = e^{-2r\Delta t} [P^2 f_{uu} + 2P(1-P) f_{ud} + (1-P)^2 f_{dd}]$$



Example: Put Option

- Consider a 2-year put option with strike price of Rs.52 on a stock whose current price is Rs.50. Suppose that there are two times of 1 year, and each time step the stock price either moves up by 20% or moves down by 20%. The risk-free rate is 5%.

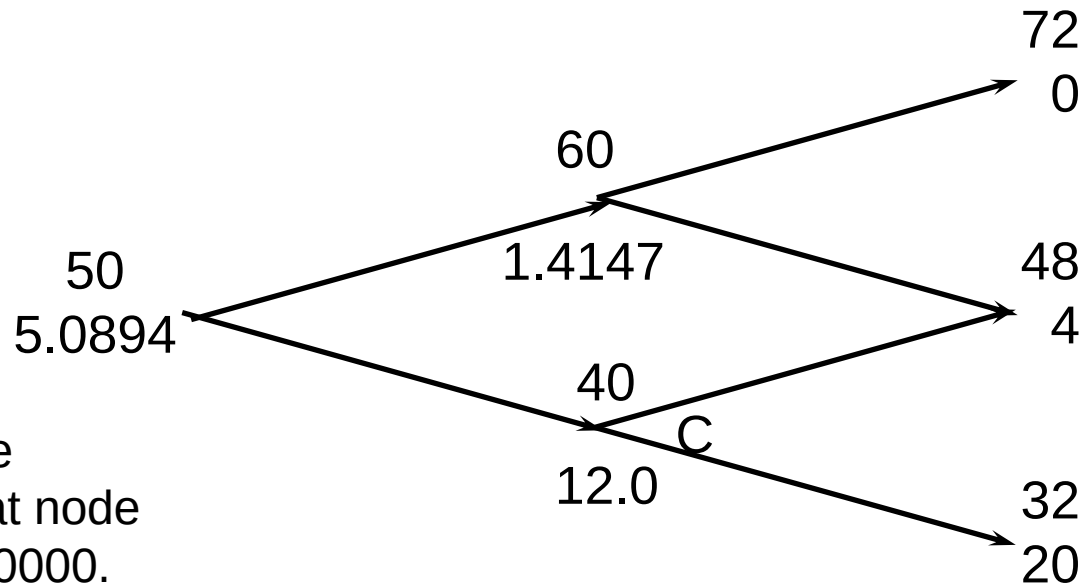
A Put Option Example



$K = 52$, time step = 1yr

$r = 5\%$, $u = 1.2$, $d = 0.8$

What Happens When the Put Option is American



The American feature increases the value at node C from 9.4636 to 12.0000.

This increases the value of the option from 4.1923 to 5.0894.

Delta

- Delta (Δ) is the ratio of the change in the price of a stock option to the change in the price of the underlying stock
- The value of Δ varies from node to node

Choosing u and d

One way of matching the volatility is to set

$$u = e^{\sigma\sqrt{\Delta t}}$$

$$d = 1/u = e^{-\sigma\sqrt{\Delta t}}$$

where σ is the volatility and Δt is the length of the time step. This is the approach used by Cox, Ross, and Rubinstein

Assets Other than Non-Dividend Paying Stocks

- For options on stock indices, currencies and futures the basic procedure for constructing the tree is the same except for the calculation of p

The Probability of an Up Move

$$p = \frac{a - d}{u - d}$$

$a = e^{r\Delta t}$ for a nondividend paying stock

$a = e^{(r-q)\Delta t}$ for a stock index where q is the dividend yield on the index

$a = e^{(r-r_f)\Delta t}$ for a currency where r_f is the foreign risk - free rate

$a = 1$ for a futures contract

Example

- Consider an American put option where the stock price is Rs.50, the strike price is Rs.52, the risk-free rate is 5%, the life of the option is 2 years, and there are two-time steps. The volatility is 30%.

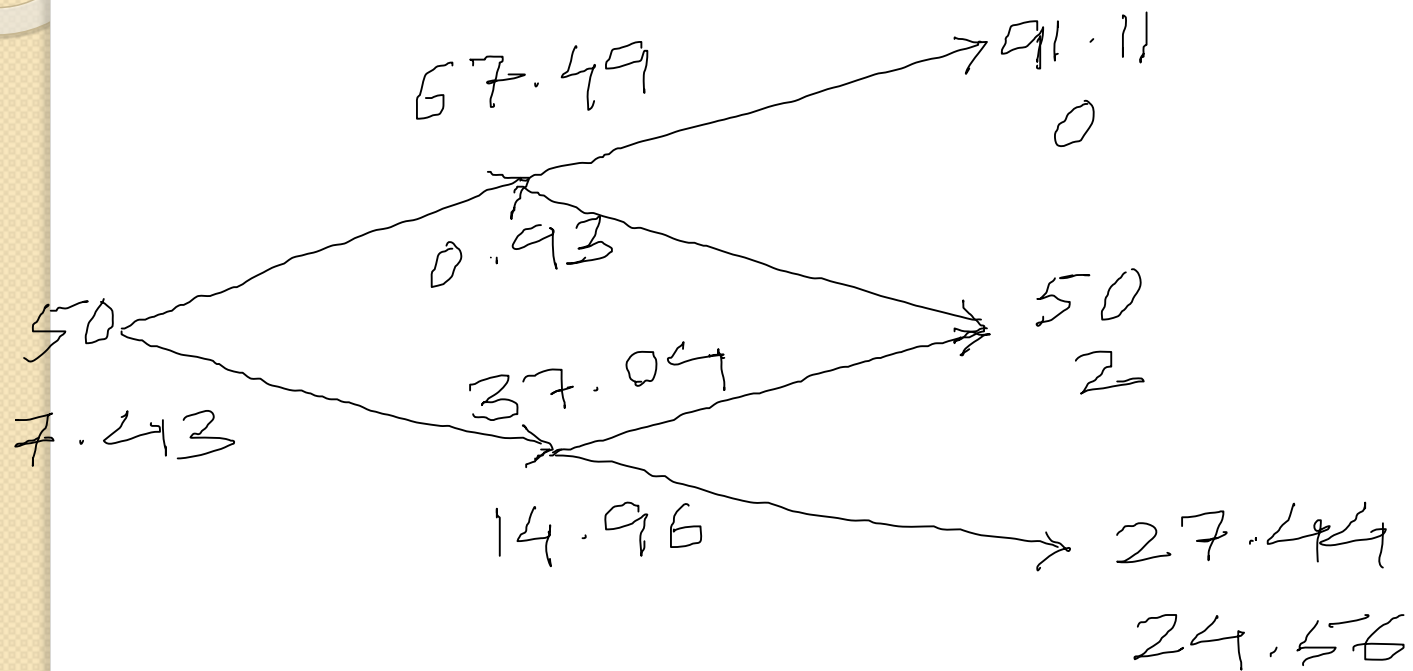
$$u = e^{\sigma\sqrt{\Delta t}}, \quad d = 1/u, \quad a = e^{r\Delta t}$$

$$p = \frac{a - d}{u - d}$$

$$u = e^{0.3 \times 1} = 1.3499$$

$$d = 0.7408, \quad a = 1.0513$$

$$p = \frac{1.0513 - 0.7408}{1.3499 - 0.7408}$$
$$= 0.5097$$



Example

A stock index is currently 810 and has a volatility of 20% and dividend yield of 2%. The risk-free rate of interest is 5%. Determine the value of the European call option with a strike price 800 using two step binomial tree.